REVISION 2021 • SEMESTER 6 • TEXTILE TECHNOLOGY

Textile Mill Management

A full-screen, syllabus-style digital handbook for Course Code 6061C. It is written as a practical textbook with module explanations, Malayalam support notes, formulas, checklists, worked cases, lab/report guidance, revision questions, model question paper and full answer key.

COURSE CODE
6061C

COURSE TITLE
Textile Mill Management

DEPARTMENT
Textile Technology

TYPE
Theory

SEMESTER
6

CATEGORY
Program Elective

USE
Study + notes PDF

FORMAT
Big handbook

6061C

Screen-filling handbook

Designed for desktop, mobile, classroom display and automatic PDF notes generation.

How to use this handbook

Read the overview first, then study each module with the diagram, formulas, practical tasks and model answers. For exam preparation, revise the question bank and answer key.

Course map	**objectives + outcomes**	**Module lessons**	**4 detailed units**
Formula bank	**calculations**	**Practice bank**	**questions + tasks**
Model paper	**official style**	**Answer key**	**full points**

Course objectives and syllabus map

Objectives

- ✓ Understand textile mill organisation and production control
- ✓ Apply maintenance, inventory and labour management basics
- ✓ Use cost and quality information for mill decision-making

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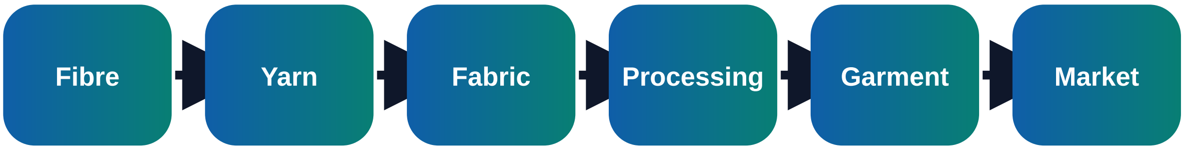
Visual process map



Module	Main outcome	Industrial focus	Expected evidence
M1	Mill organisation and functions	Organisation structure of spinning, weaving, processing and garment units; Roles of production, quality, maintenance, stores, purchase, HR and dispatch	Short notes, calculations, diagram, checklist, viva answers
M2	Production planning and control	Order planning, capacity calculation and machine allocation; Routing, scheduling, loading and line balancing	Short notes, calculations, diagram, checklist, viva answers
M3	Maintenance and inventory management	Preventive, breakdown and predictive maintenance; Machine history card, spare planning and lubrication schedule	Short notes, calculations, diagram, checklist, viva answers
M4	Cost, labour and quality management	Elements of cost: material, labour, power, overhead and waste; Labour productivity, skill matrix, training and discipline	Short notes, calculations, diagram, checklist, viva answers

Module-wise textbook

Each module is written in clear engineering language. Do not mug up headings only; understand the process flow, defects, records and decision points.

Module 1

Mill organisation and functions

Core explanation

- ✓ Organisation structure of spinning, weaving, processing and garment units
- ✓ Roles of production, quality, maintenance, stores, purchase, HR and dispatch
- ✓ Shift system, supervision, reporting and communication flow
- ✓ Productivity, quality and safety as management targets

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Exam writing method

1. Start with definition or purpose.
2. Draw block diagram/process flow wherever possible.
3. List parameters, machines/tools, defects and precautions.
4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 2

Production planning and control

Core explanation

- ✓ Order planning, capacity calculation and machine allocation
- ✓ Routing, scheduling, loading and line balancing
- ✓ Production follow-up, bottleneck identification and corrective action
- ✓ Documentation: production report, efficiency report and downtime report

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 3

Maintenance and inventory management

Core explanation

- ✓ Preventive, breakdown and predictive maintenance
- ✓ Machine history card, spare planning and lubrication schedule
- ✓ Stores layout, material issue, reorder level and ABC analysis
- ✓ Waste control, energy conservation and housekeeping

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 4

Cost, labour and quality management

Core explanation

- ✓ Elements of cost: material, labour, power, overhead and waste
- ✓ Labour productivity, skill matrix, training and discipline
- ✓ Quality cost: prevention, appraisal, internal failure and external failure
- ✓ Management information system and continuous improvement

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Exam writing method

1. Start with definition or purpose.
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3. List parameters, machines/tools, defects and precautions.
4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Formula bank and quick calculations

Formula 1

Machine Efficiency % = Actual Output / Standard Output x 100

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 2

Labour Productivity = Output / Labour Hours

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 3

Reorder Level = Maximum Consumption x Maximum Lead Time

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 4

Total Cost = Material + Labour + Overhead + Power + Waste

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Practice tasks, viva and revision

Practical / assignment tasks

- ✓ Prepare shift production report
- ✓ Calculate machine efficiency from production data
- ✓ Make ABC classification for spares
- ✓ Prepare preventive maintenance checklist

Important keywords

PPC **Efficiency** **Downtime** **Inventory** **Maintenance** **MIS** **Labour** **Cost**

Short questions

Q1.1	Explain: Organisation structure of spinning, weaving, processing and garment units	Answer with definition, process, application and one example.
Q1.2	Explain: Roles of production, quality, maintenance, stores, purchase, HR and dispatch	Answer with definition, process, application and one example.
Q1.3	Explain: Shift system, supervision, reporting and communication flow	Answer with definition, process, application and one example.
Q2.1	Explain: Order planning, capacity calculation and machine allocation	Answer with definition, process, application and one example.
Q2.2	Explain: Routing, scheduling, loading and line balancing	Answer with definition, process, application and one example.
Q2.3	Explain: Production follow-up, bottleneck identification and corrective action	Answer with definition, process, application and one example.
Q3.1	Explain: Preventive, breakdown and predictive maintenance	Answer with definition, process, application and one example.
Q3.2	Explain: Machine history card, spare planning and lubrication schedule	Answer with definition, process, application and one example.

Q3.3	Explain: Stores layout, material issue, reorder level and ABC analysis	Answer with definition, process, application and one example.
Q4.1	Explain: Elements of cost: material, labour, power, overhead and waste	Answer with definition, process, application and one example.
Q4.2	Explain: Labour productivity, skill matrix, training and discipline	Answer with definition, process, application and one example.
Q4.3	Explain: Quality cost: prevention, appraisal, internal failure and external failure	Answer with definition, process, application and one example.

Case study

A textile unit receives customer complaint related to PPC. Prepare a corrective action report using observation, data collection, root cause, corrective action, preventive action and verification.

Expected answer structure: Problem statement → data/table → cause analysis → action plan → responsible person → completion date → verification evidence.

Model question paper

Use this as a sample paper. Questions are original and arranged in diploma-style sections.

PART A - Short Answer Questions (answer all)
1. Define Textile Mill Management.
2. List four important keywords: PPC, Efficiency, Downtime, Inventory.
3. State two industrial applications of this subject.
4. Mention two common defects/problems and one preventive action.
5. Write any one important formula or calculation used in this subject.

PART B - Paragraph Questions
6. Explain the workflow of Mill organisation and functions.
7. Compare two important concepts from Production planning and control.
8. Explain the practical importance of Maintenance and inventory management in a textile unit.
9. Write the steps for documentation/reporting in this subject.

PART C - Essay / Application Questions
10. Prepare a complete process plan for Textile Mill Management in a textile industry situation.
11. Explain how quality, cost, safety and customer requirement are controlled in this subject.
12. Solve/interpret a simple industrial case using the formulas and checks given in this handbook.

Answer key and scoring points

Part A answer points

1. Give accurate definition and scope of Textile Mill Management.
2. Use keywords: PPC, Efficiency, Downtime, Inventory, Maintenance, MIS, Labour, Cost.
3. Applications must be linked to textile industry, customer requirement and quality.
4. Defects/problems should include cause and prevention.
5. Formula answers must show unit and interpretation.

Part B/C - Module 1

Mill organisation and functions

- Organisation structure of spinning, weaving, processing and garment units
- Roles of production, quality, maintenance, stores, purchase, HR and dispatch
- Shift system, supervision, reporting and communication flow
- Productivity, quality and safety as management targets

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 2

Production planning and control

- Order planning, capacity calculation and machine allocation
- Routing, scheduling, loading and line balancing
- Production follow-up, bottleneck identification and corrective action
- Documentation: production report, efficiency report and downtime report

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 3

Maintenance and inventory management

- Preventive, breakdown and predictive maintenance
- Machine history card, spare planning and lubrication schedule
- Stores layout, material issue, reorder level and ABC analysis
- Waste control, energy conservation and housekeeping

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 4

Cost, labour and quality management

- Elements of cost: material, labour, power, overhead and waste
- Labour productivity, skill matrix, training and discipline
- Quality cost: prevention, appraisal, internal failure and external failure
- Management information system and continuous improvement

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Final quick revision

Before exam

- Revise all formulas and keywords.
- Practise one diagram/process flow from every module.
- Prepare two industrial examples for every long answer.

Before lab/viva

- Know instrument/machine purpose.
- Know safety precautions and standard record format.
- Explain result, defect and correction logically.